SQL

SQL Interview Pack

8 patterns that cover most SQL interview questions.

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Swipe →

Patterns 1 to 4

- **Top N per group**
  `DENSE_RANK() OVER (PARTITION BY ...)`
- **Running total**
  `SUM() OVER (ORDER BY date)`
- **Month on month**
  `LAG(revenue) OVER (...)`
- **Remove duplicates**
  `ROW_NUMBER() ... WHERE rn = 1`

Patterns 5 to 8

- **Consecutive days**
  `date - ROW_NUMBER()`
- **Find missing rows**
  `LEFT JOIN ... WHERE b.id IS NULL`
- **Pivot without PIVOT**
  `SUM(CASE WHEN ... THEN 1 END)`
- **Employee and manager**
  `Self join on manager_id`

Your JOIN doubled the rows?

The join key is not unique on one side.

```
SELECT key, COUNT(*)
FROM t
GROUP BY key
HAVING COUNT(*) > 1;
```

Fix: remove duplicates or aggregate before you join.

Order of execution

```
FROM
JOIN
WHERE
GROUP BY
HAVING
SELECT
ORDER BY
LIMIT
```

That is why a SELECT alias fails inside WHERE.